The MEW nozzle in two dimensions

Results at the corrected feed, for the set-up of Ju, Omar, Mirihanage and Kilic

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Summary. The set-up of *Suppressing oozing during filament-based melt electrowriting* (SSRN 7582986) was simulated as a planar slice through the nozzle: PCL at 120 °C fed through a 0.4 mm nozzle at 6.5 kV, 5.5 mm above a collector moving at 35 mm s^{-1}, at the corrected feed of 0.0208 mm^3 s^{-1} (Table 1 prints 0.255 mm^3 s^{-1}). Four results:

- With the field off, the melt hangs from the nozzle as a drop that grows at the feed rate and reaches the collector after 105 to 110 s; nothing in a planar model can carry it away as a jet.
- With the field on, when the drop reaches the collector depends on how deep the model lets the melt conduct: 7 to 17 s later than with the field off for depths of 1 to 3 mm, 48 s sooner when nearly the whole drop conducts. That depth is a numerical choice, so neither the size nor the direction of the effect is a prediction.
- At 35 mm s^{-1} a straight fibre carrying the whole feed is 27.5 µm across; the reported 19 µm to 22 µm fibres carry about half of it, the fan's 9.5 µm to 13.5 µm fibres 12 to 24 %.
- The numbers are not grid-converged, and nothing after the melt reaches the collector can be used.

1 Model and set-up

Melt	PCL held at 120 °C. Viscosity 500 Pa s and surface tension 0.03 N m^{-1} are nominal values, not measurements; density 1080 kg m^{-3}.
Nozzle	0.4 mm bore, as in the paper; an assumed 1.0 mm tip face, 5.5 mm above the collector.
Field	6.5 kV. The melt conducts within 2 mm of the tip (1 and 3 mm in a test, Section 2.2) and is an insulator below, so it cannot short the gap.
Collector	35 mm s^{-1} (2100 mm min^{-1}), as in every experiment.
Feed	0.0208 mm^3 s^{-1}, Table 1's corrected value (Murat Kilic, 3 November), with and without the fan; the melt leaves the bore at 0.17 mm s^{-1}. Earlier readings: Section 2.4.
Grid	25 µm cells over an 8×6.8 mm slice, 14 µs steps, and 50 µm cells as a check. A free-surface lattice Boltzmann model of the melt (D2Q9, central moments; the air is not simulated) and Laplace's equation for the potential in the air (D2Q5); the melt surface feels the pull $\varepsilon_0 E^2/2$.

Matched to the experiment: the capillary numbers of the feed (2.8) and of the collector (583), the Bond number (0.014), and the ratio of electric to capillary pressure, $\varepsilon_0 V^2/\gamma R = 62$ at 6.5 kV. Not matched: inertia, which is negligible in the experiment and kept small in the model (Reynolds number at most 0.32).

2 Results

2.1 Field off: a drop that reaches the collector in about two minutes

The capillary number in the bore is 2.8. The melt hangs from the nozzle as a drop that grows at the feed rate; its lowest point falls at about 0.04 mm s^{-1} and reaches the collector after 105.0 s on 25 µm cells and 110.3 s on 50 µm cells (Fig. 1), 22 times later than the printed feed's column. The melt within 1 mm of the tip settles at 0.69 mm^2, about what the printed feed piles there. A planar sheet carrying this feed at 35 mm s^{-1} would be 1.9 µm thick, far below the cells, so nothing carries melt away and no jet forms. In three dimensions a drop of the same outline needs more melt, by about its width over the bore's, four to five times here, so a round nozzle would take longer to reach the collector: an estimate, not a run.

2.2 Field on: later or sooner, depending on a choice in the model

With the field on the drop first falls faster, 0.40 mm (25 µm cells) and 0.45 mm (50 µm) below the field-off drop at 20 s. Then it slows and widens (2.05 to 2.10 mm at 100 s, against 1.90 mm) and is still 0.09 and 0.075 mm above the collector when the runs end at 120 s; extrapolated, it touches about 18 and 11 s after the field-off drop. The slowdown comes exactly when the drop's lowest point leaves the zone where the model lets the melt conduct, 2 mm below the tip: its speed falls there from 0.090 to 0.034 mm s^{-1} (50 µm), while the field-off drop passes the same height unchanged at 0.05 mm s^{-1}. The model needs that limit, or the melt would short the gap, but nothing in it fixes the depth. With the limit at 1 or 3 mm instead (Fig. 2a) the bend moves with it, the runs are identical to every printed digit until the drop first passes the shallower limit, and the drop reaches the collector 7.4, about 11 and 17.0 s after the field-off drop for 1, 2 and 3 mm.

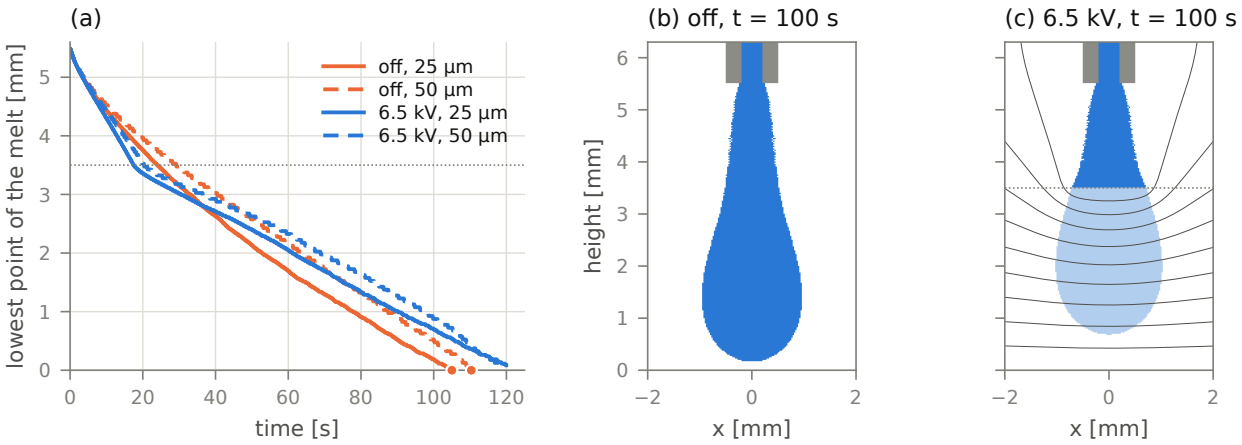


Figure 1. The corrected feed, 0.0208 $mm^3\ s^{-1}$. (a) Height of the lowest point of the melt against time with the field off (orange) and on (blue), on 25 µm (solid) and 50 µm cells (dashed). Dots mark first contact with the collector, which neither field-on run reaches by 120 s; dotted, the depth below which the melt does not conduct. (b, c) The drop at 100 s on 25 µm cells, field off and on; in (c) the melt that conducts is dark and the melt below the limit light, with equipotentials every 10 % of the voltage.

While the whole drop conducts, its strongest pull is at its lowest point and draws it down. Once it hangs below the limit, its conducting part ends in a flat face inside it, and the strongest pull moves to that face's edge on the drop's sides, 7 to 56 times the typical pull on the conducting surface (Fig. 2b). That edge is numerical. With the limit at 5 mm, 0.5 mm above the collector, the whole drop conducts until it is nearly down, as a melt whose charge relaxes in much less than its two minutes of growth would. The field then draws it into a column that runs away toward the collector, 2.5 to 0.5 mm in 5.5 s, and it touches 48 s *before* the field-off drop, with 9 to 14 kV mm^{-1} in the last millimetre, well above the roughly 3 kV mm^{-1} at which air breaks down. So neither the size of the field's effect nor its direction is a prediction.

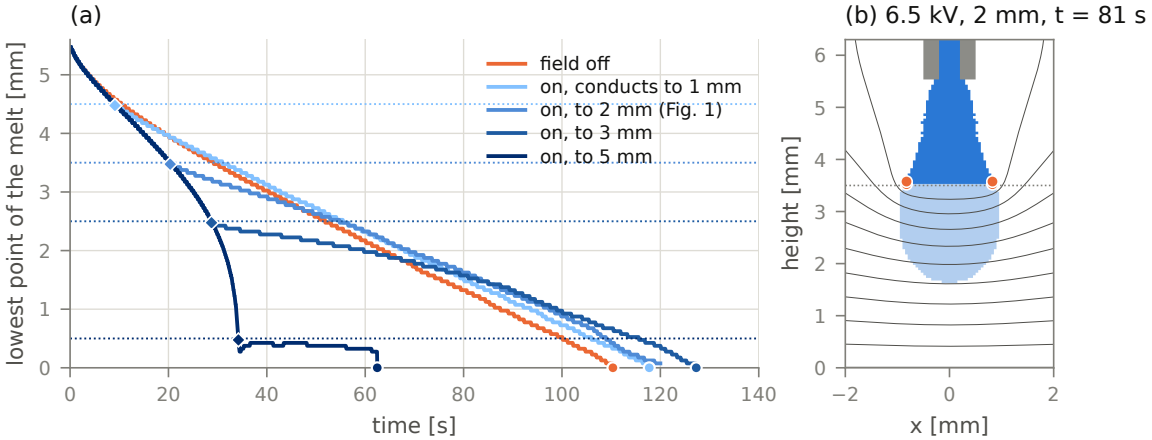


Figure 2. How deep the melt may conduct, on 50 µm cells. (a) Height of the lowest point of the melt against time with the field off (orange) and on, the melt conducting to 1, 2, 3 or 5 mm below the tip (light to dark blue; 2 mm is the default of Fig. 1). Dotted, each depth; diamonds, the drop's lowest point passing it; dots, first contact. (b) The 2 mm run at 81 s: melt that conducts (dark) and melt that does not (light), equipotentials every 10 % of the voltage, and the four surface cells pulled hardest (orange), at the edge of the conducting zone.

2.3 What the fibres carry

A straight fibre laid at 35 mm s^{-1} carries its cross-section times that speed; carrying the whole feed it would be 27.5 µm across. The paper's 19 µm to 22 µm fibres (after nozzle cleaning) carry 48 to 64 % of the feed, its 9.5 µm to 13.5 µm fibres (with the fan) 12 to 24 %. A Taylor cone and a lag angle thin the jet, but stretching conserves volume: a 20.5 µm fibre carrying the whole feed would have to be laid at 63 mm s^{-1}, an 11.5 µm one at 200 mm s^{-1}, and a fibre laid faster than the collector moves coils rather than lying straight. So if the fibres are straight, part of the feed does not reach the collector (question 1).

2.4 The two earlier readings of the feed

At the printed 0.255 $mm^3\ s^{-1}$, twelve times the corrected feed (capillary number 34), the melt is pushed out as a column, field on or off, and reaches the collector at 4.7 s on 25 µm cells; at 0.255 $mm^3\ min^{-1}$ (0.56) the field

shapes a meniscus that grows at exactly the feed rate, at 6.5 and 12 kV alike. The first version of this note, of 3 November, has both.

2.5 How the numbers depend on the cell size

Nothing converges, and the nozzle is why. With no wetting the model treats the nozzle as a surface the melt does not wet, so the melt leaves from the edge of the bore and creeps exactly two cells onto the tip face, whatever a cell measures: the neck is 0.60, 0.50 and 0.45 mm wide on 50, 25 and 12.5 µm cells, and the column follows it. At the printed feed first contact came at 5.8, 4.7 and 4.0 s on the three grids; at the corrected feed the field-off drop touches at 110.3 and 105.0 s on 50 and 25 µm cells, with 0.85 and 0.69 mm^2 of melt within 1 mm of the tip. A wetting melt would hang from the nozzle's outer edge, 1.0 mm wide, a bigger change than any grid makes. At about two days a run, 12.5 µm cells at the corrected feed are beyond a laptop.

2.6 After the melt reaches the collector

Nothing after contact can be used at the corrected feed. The collector drags the drop to the side of the 8 mm box within 2 s; on 50 µm cells its neck snaps and 41 % of the melt drains out of the box, while on 25 µm cells it lifts off and hangs from the nozzle as a hook. At the printed feed, a difference between field on and off after contact on 25 µm cells was gone on 12.5 µm cells.

3 What the model leaves out

- **Three dimensions.** A planar meniscus has no conical equilibrium, the jet is far below the grid, a flat tip concentrates the field less than a round one, and a planar drop holds less melt for its outline.
- **Wetting.** The nozzle and the collector are surfaces the melt does not wet; a contact-angle law since added spreads the melt by only one cell a side in 4 s, and its contact line sticks, more so on finer grids.
- **Charge and conduction.** A perfect conductor down to a chosen depth and an insulator below, with no free charge and no current.
- **Heat and rheology.** Isothermal at 120 °C (the hot end, where the fan acts, was modelled separately) and Newtonian at a nominal 500 Pa s.

4 How far to trust it

Electric pull, against exact solutions	Its pattern is right to 0.05 %, a moving surface's response to 3 % once the surface's own stiffness is allowed for: deflections of one or two cells come out 9 % to 25 % short.
Surface tension, Laplace's law	Right to 1.2 % for drops of 12 cells or more, 3.4 % at 8 cells; the bore is 8 cells from wall to axis.
Three cell sizes, 50, 25 and 12.5 µm	Not converged, all set by the neck the melt leaves through (Section 2.5).
How deep the melt conducts	Sets the field's effect: the drop reaches the collector 7.4, about 11 and 17.0 s after the field-off drop for depths of 1, 2 and 3 mm, and 48 s before it for 5 mm.
Mass	Conserved to 0.3 % until contact (0.6 % at the printed feed); after contact off by up to 3.3 %, and by up to 5.4 % while a thread is drawn thin at the printed feed, which is not yet explained.
Symmetry	Held mirror-symmetric until contact, as the set-up is; otherwise round-off pushes the column up to 0.4 mm sideways.

5 Questions for the experiment

1. Where does the rest of the feed go? The fibres carry 12 to 64 % of 0.0208 $mm^3\ s^{-1}$ at 35 $mm\ s^{-1}$; weighing the deposit over a timed run, against 1.25 mm^3 (about 1.4 mg) fed a minute, would show whether all of it reaches the collector.
2. How well does this PCL conduct at 120 °C? Its conductivity and permittivity would let the model replace the conducting depth that now decides what the field does.
3. Are there measured values of its viscosity, ideally a flow curve, and surface tension? The model uses 500 Pa s and 0.03 $N\ m^{-1}$.
4. Does the melt wet the nozzle tip, hanging from the bore or from its outer edge? That choice moves the melt more than any grid here.

M3LB (github.com/alexderosis/M3LB, branch `electrowriting`), double precision: `demonstrator/mew_nozzle -feed 0.0208`, with `-nofield`, and `-lc 1`, 3 or 5 for the conducting depth; cells `-R 8` (the default) and `-R 4 -Ucol 0.04`. The earlier readings: `-feed s` or `min`, `-V 12`, `-R 16 -Ucol 0.01`. Figures: `paper/mew_nozzle/make_figs.py`.